

23. To Execute Large Pipelines, Launch Multiple Jobs, Measure Stability, Monitor Resources, Detect Failures, Analyze Performance.

Code:

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <sys/resource.h>
#include <string.h>

#define PIPELINE_COUNT 3

/* -----
Calculate elapsed time
----- */
double getTimeDifference(struct timeval start,
                        struct timeval end) {
    double seconds =
        (double)(end.tv_sec - start.tv_sec);
    double microseconds =
        (double)(end.tv_usec - start.tv_usec) / 1000000.0;
    return seconds + microseconds;
}

/* -----
Run one 3-stage pipeline
----- */
int runPipeline(int jobNumber) {
    int pipe1[2];
    int pipe2[2];
    pid_t p1, p2, p3;
    printf("\n[Job %d] Starting pipeline...\n",
        jobNumber);

    /* Create first pipe.
    */
    if (pipe(pipe1) == -1) {
        perror("pipe1");
        return 0;
    }

    /* Create second pipe.
    */
    if (pipe(pipe2) == -1) {
        perror("pipe2");
        close(pipe1[0]);
        close(pipe1[1]);
        return 0;
    }
}
```

```
/* -----
Process 1
seq 1 1000
----- */

p1 = fork();
if (p1 == -1) {
    perror("fork process 1");
    return 0;
}

if (p1 == 0) {
    /* stdout -> pipe1
    */
    dup2(pipe1[1], STDOUT_FILENO);

    close(pipe1[0]);
    close(pipe1[1]);

    close(pipe2[0]);
    close(pipe2[1]);

    execlp("seq", "seq", "1", "1000", NULL);
    perror("seq failed");
    exit(1);
}

/* -----
Process 2
sort
----- */

p2 = fork();
if (p2 == -1) {
    perror("fork process 2");
    return 0;
}

if (p2 == 0) {
    /* stdin <- pipe1
    */
    dup2(pipe1[0], STDIN_FILENO);

    /* stdout -> pipe2
    */
    dup2(pipe2[1], STDOUT_FILENO);

    close(pipe1[0]);
}
```

```

        /* stdin <- pipe1
        */
        dup2(pipe1[0], STDIN_FILENO);

        /*
        * stdout -> pipe2
        */
        dup2(pipe2[1], STDOUT_FILENO);

        close(pipe1[0]);
        close(pipe1[1]);

        close(pipe2[0]);
        close(pipe2[1]);

        execlp("sort", "sort", NULL);

        perror("sort failed");
        exit(1);
    }

    /* -----
    Process 3
    wc -l
    ----- */

    p3 = fork();
    if (p3 == -1) {
        perror("fork process 3");
        return 0;
    }

    if (p3 == 0) {
        /*
        * stdin <- pipe2
        */
        dup2(pipe2[0], STDIN_FILENO);

        close(pipe1[0]);
        close(pipe1[1]);

        close(pipe2[0]);
        close(pipe2[1]);

        /*
        * Final output goes to terminal.
        */
        execlp("wc", "wc", "-l", NULL);

        perror("wc failed");
        exit(1);
    }

    /* -----
    Parent closes all pipe descriptors
    -----

```

```

    /* -----
    Parent closes all pipe descriptors
    ----- */

    close(pipe1[0]);
    close(pipe1[1]);

    close(pipe2[0]);
    close(pipe2[1]);

    /* -----
    Wait for all pipeline processes
    ----- */

    int status1;
    int status2;
    int status3;

    waitpid(p1, &status1, 0);
    waitpid(p2, &status2, 0);
    waitpid(p3, &status3, 0);

    /*
    * Detect failures.
    */
    if (!WIFEXITED(status1) ||
        WEXITSTATUS(status1) != 0) {
        printf("[Job %d] Process 1 failed.\n",
            jobNumber);

        return 0;
    }

    if (!WIFEXITED(status2) ||
        WEXITSTATUS(status2) != 0) {
        printf("[Job %d] Process 2 failed.\n",
            jobNumber);

        return 0;
    }

    if (!WIFEXITED(status3) ||
        WEXITSTATUS(status3) != 0) {
        printf("[Job %d] Process 3 failed.\n",
            jobNumber);

        return 0;
    }

    printf("[Job %d] Pipeline completed successfully.\n",
        jobNumber);

    return 1;
}
|

```

```

/* -----
Main
----- */

int main() {
    struct timeval start;
    struct timeval end;
    struct rusage usage;
    int successfulJobs = 0;
    int failedJobs = 0;

    printf("Large Pipeline Performance Monitor\n");
    printf("-----\n");

    printf("Pipeline:\n");
    printf("seq 1 1000 | sort | wc -l\n");

    printf("\nLaunching %d pipeline jobs...\n",
        PIPELINE_COUNT);

    /*
    * Start performance timer.
    */
    gettimeofday(&start, NULL);

    /* -----
    Launch multiple pipeline jobs
    ----- */

    for (int i = 1; i <= PIPELINE_COUNT; i++) {
        if (runPipeline(i)) {
            successfulJobs++;
        } else {
            failedJobs++;
        }
    }

    /*
    * Stop performance timer.
    */
    gettimeofday(&end, NULL);

    /*
    * Get resource usage of child processes.
    */
    if (getrusage(RUSAGE_CHILDREN, &usage) == -1) {

```

```

/*
 * Get resource usage of child processes.
 */
    if (getrusage(RUSAGE_CHILDREN, &usage) == -1) {
        perror("getrusage");
        return 1;
    }

    double elapsed =
        getTimeDifference(start, end);

    /* -----
    Performance Report
    ----- */

    printf("\n===== \n");
    printf("      PERFORMANCE REPORT\n");
    printf("===== \n");

    printf("Total Jobs      : %d\n",
        PIPELINE_COUNT);

    printf("Successful Jobs : %d\n",
        successfulJobs);

    printf("Failed Jobs     : %d\n",
        failedJobs);

    printf("Execution Time  : %.6f seconds\n",
        elapsed);

    printf("User CPU Time   : %ld.%06ld seconds\n",
        usage.ru_utime.tv_sec,
        usage.ru_utime.tv_usec);

    printf("System CPU Time : %ld.%06ld seconds\n",
        usage.ru_stime.tv_sec,
        usage.ru_stime.tv_usec);

    printf("Maximum Memory  : %ld KB\n",
        usage.ru_maxrss);

    printf("===== \n");

    /*
    * Stability check.
    */
    if (failedJobs == 0) {
        printf("Stability Test   : PASSED\n");
        printf("All pipelines completed successfully.\n");
    } else {
        printf("Stability Test   : FAILED\n");

```

```

        usage.ru_stime.tv_sec,
        usage.ru_stime.tv_usec);

printf("Maximum Memory   : %ld KB\n",
       usage.ru_maxrss);

printf("=====\n");

/*
 * Stability check.
 */
if (failedJobs == 0) {

    printf("Stability Test   : PASSED\n");
    printf("All pipelines completed successfully.\n")

} else {

    printf("Stability Test   : FAILED\n");
    printf("Some pipeline jobs failed.\n");

}

/*
 * Performance analysis.
 */
if (elapsed < 1.0) {

    printf("Performance       : GOOD\n");

} else {

    printf("Performance       : REVIEW REQUIRED\n");

}

printf("=====\n");

return 0;
}

```

Result:

```

saanvi_kompelli@DESKTOP-2U40RG7: L$ nano Skill23a.c
saanvi_kompelli@DESKTOP-2U40RG7: L$ nano Skill23a.c
saanvi_kompelli@DESKTOP-2U40RG7: L$ gcc Skill23a.c -o s23a
saanvi_kompelli@DESKTOP-2U40RG7: L$ ./s23a
Large Pipeline Performance Monitor

Pipeline:
seq 1 1000 | sort | wc -l

Launching 3 pipeline jobs...

[Job 1] Starting pipeline...
1000
[Job 1] Pipeline completed successfully.

[Job 2] Starting pipeline...
1000
[Job 2] Pipeline completed successfully.

[Job 3] Starting pipeline...
1000
[Job 3] Pipeline completed successfully.

=====
PERFORMANCE REPORT
=====
Total Jobs       : 3
Successful Jobs  : 3
Failed Jobs      : 0
Execution Time   : 0.021224 seconds
User CPU Time    : 0.015458 seconds
System CPU Time  : 0.016926 seconds
Maximum Memory   : 8540 KB
=====
Stability Test   : PASSED
All pipelines completed successfully.
Performance      : GOOD
=====
saanvi_kompelli@DESKTOP-2U40RG7: L$ |

```

To Document Architecture, Create Flow Diagrams, Explain Execution Paths, Describe Syscalls, Record Design Decisions, Review Documentation.

Code:

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

/*
 * SKILL 24:
 * Architecture Documentation and System Call Demonstration
 *
 * Architecture:
 *
 *      MAIN PROCESS
 *      |
 *      v
 *      CHILD PROCESS
 *      |
 *      v
 *      COMMAND EXECUTION
 *      |
 *      v
 *      COMMAND OUTPUT
 *
 * Parent waits using waitpid()
 */

void documentArchitecture() {
    printf("\n");
    printf("=====SYSTEM ARCHITECTURE\n");
    printf("=====");

    printf("1. Main Process\n");
    printf("    |");
    printf("    | fork()\n");
    printf("    v");
    printf("2. Child Process\n");
    printf("    |");
    printf("    | execvp()\n");
    printf("    v");
    printf("3. Command Execution\n");
    printf("    |");
    printf("    v");
    printf("4. Command Output\n");
    printf("    |");
    printf("    v");
    printf("5. Parent waits using waitpid()\n");
}

void executeCommand() {
    pid_t pid;
    int status;

    printf("\n");
    printf("=====EXECUTION PATH\n");
    printf("=====");

    printf("Parent: Creating child process...\n");

    /*
     * System Call 1: fork()
     */
}
```

```
/*
 * System Call 1: fork()
 */
pid = fork();

if (pid < 0) {
    perror("fork");
    return;
}

/*
 * CHILD PROCESS
 */
if (pid == 0) {
    printf("Child : Process created successfully.\n");
    printf("Child : PID = %d\n", getpid());
    printf("Child : Executing 'ls' command...\n");

    /*
     * System Call 2:
     * execvp()
     */
    char *args[] = {"ls", "-l", NULL};
    execvp(args[0], args);

    /*
     * execvp() only returns
     * when execution fails.
     */
    perror("execvp");
    exit(1);
}

/*
 * PARENT PROCESS
 */
else {
    printf("Parent: Child PID = %d\n", pid);
    printf("Parent: Waiting for child...\n");

    /*
     * System Call 3:
     * waitpid()
     */
    if (waitpid(pid, &status, 0) == -1) {
        perror("waitpid");
        return;
    }

    if (WIFEXITED(status)) {
        printf("Parent: Child exited normally.\n");
        printf("Parent: Exit status = %d\n",
            WEXITSTATUS(status));
    }
}
```

```

        if (WIFEXITED(status)) {
            printf("Parent: Child exited normally.\n");
            printf("Parent: Exit status = %d\n",
                WEXITSTATUS(status));
        } else {
            printf("Parent: Child did not exit normally.\n");
        }
    }
}

void documentSystemCalls() {
    printf("\n===== \n");
    printf("          SYSTEM CALLS USED\n");
    printf("===== \n");
    printf("fork()    -> Creates a child process\n");
    printf("execvp()  -> Executes a new program\n");
    printf("waitpid() -> Parent waits for child\n");
    printf("getpid()  -> Retrieves process ID\n");
}

void documentDesignDecisions() {
    printf("\n===== \n");
    printf("          DESIGN DECISIONS\n");
    printf("===== \n");
    printf("1. Parent creates a child using fork().\n");
    printf("2. Child executes the command using execvp().\n");
    printf("3. Parent waits using waitpid().\n");
    printf("4. Exit status is checked for reliability.\n");
    printf("5. Functions are separated for clarity.\n");
}

int main() {
    printf("SYSTEM ARCHITECTURE DOCUMENTATION DEMO\n");
    printf("===== \n");
    documentArchitecture();
    documentSystemCalls();
    documentDesignDecisions();
    executeCommand();

    printf("\n===== \n");
    printf("          DOCUMENTATION REVIEW\n");
    printf("===== \n");
    printf("Architecture : Documented\n");
    printf("Flow         : Explained\n");
    printf("System Calls : Identified\n");
    printf("Execution    : Verified\n");
    printf("Design       : Recorded\n");

    printf("\nDocumentation review completed successfully.\n");
}

```

```

    printf("Design       : Recorded\n");
    printf("\nDocumentation review completed successfully.\n");

    return 0;
}

```

Result:

```

saanvi@konnelli@DESKTOP-2U40RG7: $ nano Skill23b.c
saanvi@konnelli@DESKTOP-2U40RG7: $ gcc Skill23b.c -o s23b
saanvi@konnelli@DESKTOP-2U40RG7: $ ./s23b
SYSTEM ARCHITECTURE DOCUMENTATION DEMO
=====
          SYSTEM ARCHITECTURE
=====
1. Main Process
   |
   v
2. Child Process
   |
   v
3. Command Execution
   |
   v
4. Command Output
   |
   v
5. Parent waits using waitpid()
=====
          SYSTEM CALLS USED
=====
fork()    -> Creates a child process
execvp()  -> Executes a new program
waitpid() -> Parent waits for child
getpid()  -> Retrieves process ID
=====
          DESIGN DECISIONS
=====
1. Parent creates a child using fork().
2. Child executes the command using execvp().
3. Parent waits using waitpid().
4. Exit status is checked for reliability.
5. Functions are separated for clarity.
=====
          EXECUTION PATH
=====
Parent: Creating child process...
Parent: Child PID = 4295
Parent: Waiting for child...
Child : Process created successfully.
Child : PID = 4295
Child : Executing 'ls' command...
total 968
-rwxr-xr-x 1 saanvi_kompelli@DE:saanvi_kompelli@DESKTOP-2U40RG7: 3 16:19 S2a
-rwxr-xr-x 1 saanvi_kompelli@DE:saanvi_kompelli@DESKTOP-2U40RG7: 3 16:24 S2b
-rwxr-xr-x 1 ketana_sindhu ketana_sindhu 16776 Oct 3 16:35 S3a
-rwxr-xr-x 1 ketana_sindhu saanvi_kompelli@DESKTOP-2U40RG7: 3 16:41 S3b
-rw-r--r-- 1 saanvi_kompelli@DESKTOP-2U40RG7: 3 17:18 Skill6
-rw-r--r-- 1 saanvi_kompelli@DESKTOP-2U40RG7: 3 17:18 Skill1
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 316 Aug 24 06:01 Skill1
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 2695 Oct 4 03:04 Skill1

```

```
-rwxr-xr-x 1 ketana_sindhu ketana_sindhu 16480 Oct 4 08:44 s23b
-rwxr-xr-x 1 saanvi_kompelli@DE!saanvi_kompelli@DESKTOP-2U40RG7: } 16:54 s4a
-rwxr-xr-x 1 ketana_sindhu ketana_sindhu 16640 Oct 3 16:59 s4b
-rwxr-xr-x 1 saanvi_kompelli@DESKTOP-2U40RG7: \u 16336 Oct 3 17:05 s5a
-rwxr-xr-x 1 ketana_sindhu saanvi_kompelli@DESKTOP-2U40RG7: } 17:11 s5b
-rwxr-xr-x 1 saanvi_kompelli@DESKTOP-2U40RG7: \u 16336 Oct 3 17:22 s6a
-rwxr-xr-x 1 ketana_sindhu saanvi_kompelli@DESKTOP-2U40RG7: } 17:25 s6b
-rwxr-xr-x 1 saanvi_kompelli@DESKTOP-2U40RG7: \u 16304 Oct 3 17:28 s7a
-rwxr-xr-x 1 ketana_sindhu ketana_sindhu 16488 Oct 3 17:32 s7b
-rwxr-xr-x 1 saanvi_kompelli@DE!saanvi_kompelli@DESKTOP-2U40RG7: } 17:36 s8a
-rwxr-xr-x 1 ketana_sindhu ketana_sindhu 16992 Oct 3 17:39 s8b
-rwxr-xr-x 1 saanvi_kompelli@DE!saanvi_kompelli@DESKTOP-2U40RG7: \u 02:53 s9a
-rwxr-xr-x 1 ketana_sindhu ketana_sindhu 16992 Oct 4 02:59 s9b
-rwxr-xr-x 1 saanvi_kompelli@DE!saanvi_kompelli@DESKTOP-2U40RG7: } 03:55 sh
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 4968 Sep 29 03:55 shell_history.c
-rwxr-xr-x 1 saanvi_kompelli@DESKTOP-2U40RG7: \u 16176 Aug 24 05:55 skill
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 3639 Oct 3 16:54 skill4_tokenizer.c
Parent: Child exited normally.
Parent: Exit status = 0
```

```
=====
DOCUMENTATION REVIEW
=====
Architecture : Documented
Flow          : Explained
System Calls  : Identified
Execution     : Verified
Design        : Recorded
```

```
Documentation review completed successfully.
saanvi_kompelli@DESKTOP-2U40RG7: $SKILL$ |
```